

models (GLMs), random forests, and decision trees in R.

The problem statement

The diagnosis of heart disease is a difficult task, but automation leads to better predictions about a patient's heart condition. Data relating to resting blood pressure, cholesterol, age, sex, type of chest pain, fasting blood sugar, ST depression, and exercise-induced angina can all help to predict the likelihood of having a heart attack. Decision trees, random forests, and GLMs can be trained on the given dataset and be used to view the predicted classes, where 0 equals less chances and 1 equals more chances of a heart attack.

Setting up the health monitoring system

- Import the packages Rplot, RColorBrewer, Rattle, and randomForest.
- Download and read the dataset from Kaggle at <https://www.kaggle.com/datasets/nareshbhat/health-care-data-set-on-heart-attack-possibility>.
- View the statistics of the variables in the dataset using the function 'summary'.
- Analyse the data, specific to resting blood pressure.
- Using the 'cor' function, find the correlation between resting blood pressure and age.
- Construct a logistic regression model using a GLM and view the output plots.
- Encode the target values into categorical values.
- Split the dataset into training and testing data in the ratio 70:30.
- Construct a decision tree model.
- Target variable is categorised based on resting blood pressure, serum cholesterol, and maximum heart rate achieved.

Installing the necessary packages

```
In [1]: install.packages("rpart.plot")
install.packages("rattle")
install.packages("randomForest")
```

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependencies 'bitops', 'XML'

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Loading all the necessary Libraries

```
In [2]: library(rpart) #used for building classification and regression trees.
library(rpart.plot)
library(RColorBrewer) # help you choose sensible colour schemes for figures
library(rattle) # provides a collection of utilities functions for a data
library(randomForest) #has the function randomForest() which is used to c
```

Loading required package: tibble

Loading required package: bitops

Rattle: A free graphical interface for data science with R.
Version 5.5.1 Copyright (c) 2006-2021 Togaware Pty Ltd.
Type 'rattle()' to shake, rattle, and roll your data.

randomForest 4.7-1

Type rfNews() to see new features/changes/bug fixes.

Attaching package: 'randomForest'

The following object is masked from 'package:rattle':

importance

Fig. 1: Installing and loading libraries

- Plot the decision tree and view the output.
- Devise a random forest model based on the relationship between resting blood pressure, old peak blood pressure, and chest pain type.
- View the confusion matrix and importance of each predictor.

The required data is fetched from the patient through IoT-based smart wearables and is processed using the R tool. We will be using R version 3.3.2 (2016-10-31) and RStudio version 1.2.1335 here.

Now, load all the necessary libraries:

```
library(rpart) #used for building
classification and regression trees.
library(rpart.plot)
library(RColorBrewer) # help you choose
sensible colour schemes for figures
library(rattle) # provides a collection
of utilities functions for a data
scientist. library(randomForest) #Used
to create and analyse random forests.
```

Fig. 1 shows the installation of all the necessary packages and libraries.